

IT SUPPORT TROUBLESHOOTING MANUAL

Phase 08: Documentation

Structured First-Line Troubleshooting Reference

Document ID: P07-DOC-TS-001 | **Version:** 1.0 | **Owner:** IT Support Team | **Classification:** Internal Use

1. Troubleshooting Philosophy

Troubleshooting should be systematic, evidence-based, and proportionate to the business impact. The technician should identify the symptom, determine scope, establish a likely cause, test the hypothesis, apply the least disruptive approved correction, verify the result, and document the outcome.

2. Standard Troubleshooting Lifecycle

Identify → Assess Impact → Gather Evidence → Isolate → Hypothesize → Test → Correct → Verify → Document → Resolve or Escalate

Stage	Objective	Example Output
Identify	Understand the reported symptom	Clear problem statement
Assess	Determine urgency and scope	Priority and impact
Gather	Collect relevant evidence	Errors, logs, status
Isolate	Narrow the possible cause	Affected component
Hypothesize	Form a testable explanation	Likely cause
Test	Validate the hypothesis	Test result
Correct	Apply approved fix	Configuration/action
Verify	Confirm service restoration	Successful functional test
Document	Record what happened	Ticket notes
Escalate	Transfer unresolved work	Complete escalation package

3. Technician Safety and Security Rules

- Use only authorized systems and accounts.
- Do not request or record user passwords, MFA codes, private keys, or recovery secrets.
- Do not disable security controls merely to complete troubleshooting.
- Protect business and personal information encountered during support.
- Use approved administrative procedures for privileged changes.
- Do not delete logs or evidence unless the approved procedure explicitly requires it.

- Confirm the target device before making changes.
- Document changes that could affect system availability.

4. Initial Assessment

1. Record the user's exact symptom in plain language.
2. Identify the affected device, application, service, or resource.
3. Determine when the issue began.
4. Determine whether the issue is intermittent or persistent.
5. Ask whether anything changed immediately before the problem.
6. Determine whether one user, a group, a site, or multiple sites are affected.
7. Assess business impact and confirm ticket priority.

Question	Why It Matters
What exactly is failing?	Defines the symptom.
When did it start?	Helps correlate changes/events.
Who is affected?	Determines scope.
What changed?	Identifies likely triggers.
Is there a workaround?	Helps assess business impact.
What was already tried?	Prevents duplicated or harmful actions.

5. Evidence Collection

- Capture exact error texts where useful.
- Record relevant timestamps.
- Check approved system/application logs.
- Record service status and configuration observations.
- Use screenshots only when they materially support diagnosis.
- Sanitize sensitive information before sharing evidence.
- Keep evidence associated with the correct ticket.

Evidence Quality

Good evidence is relevant, time-stamped, understandable to another technician, and sufficient to support the conclusion. Avoid collecting unrelated system information.

6. Hardware Troubleshooting

Common Symptoms

- Device will not power on
- No display
- Peripheral not detected
- Overheating or abnormal noise
- Intermittent hardware failure
- Battery/charging problem

Procedure

1. Confirm power source and visible indicators.
2. Check approved cables, ports, and physical connections.
3. Check for visible damage or unusual hardware behavior.
4. Determine whether the issue follows the peripheral or remains with the device.
5. Use an approved known-good component for controlled comparison when appropriate.
6. Run approved diagnostics if available.
7. Document findings and escalate suspected hardware failure.

Escalate When

- Physical damage
- Suspected component failure
- Repeated hardware diagnostics failure
- Potential data-loss condition
- Warranty or replacement required

7. Operating System Troubleshooting

Procedure

1. Record the symptoms and recent changes.
2. Check CPU, memory, disk, and active-process conditions.
3. Check available storage.
4. Review relevant system events/logs.
5. Confirm required services are running.
6. Apply approved updates or configuration corrections.
7. Restart only when appropriate and after protecting user work.
8. Verify the affected function.

8. Application Troubleshooting

1. Identify application name and version.
2. Capture the exact error.
3. Determine whether the problem affects one user or multiple users.
4. Check whether other applications work normally.
5. Restart the application when appropriate.
6. Review approved application logs.
7. Check service, license, network, or dependency status when relevant.
8. Repair/update/reinstall only through an approved procedure.
9. Verify normal operation.

Common Application Symptoms

Symptom	Initial Checks
Application freezes	Resource usage, recent changes, application restart
Application crashes	Error message, logs, version, recent updates
Cannot sign in	Account status, network, authentication method

Cannot save	Storage, permissions, application path
Slow application	Resource usage, network/service dependencies

9. Network Troubleshooting

1. Determine whether the issue is isolated or widespread.
2. Check physical Ethernet or approved wireless connectivity.
3. Review endpoint IP address, gateway, and DNS configuration.
4. Test connectivity to the local gateway where appropriate.
5. Test an approved internal destination.
6. Test DNS resolution when relevant.
7. Check whether a known network service or infrastructure component is unavailable.
8. Document test results and escalate infrastructure issues.

Example Authorized Commands

Purpose	Example
Linux addressing	ip addr
Linux routes	ip route
DNS status	resolvectl status
Connectivity	ping <approved-destination>
DNS lookup	nslookup <approved-name>
Path test	tracert <approved-destination>

Replace placeholders with approved destinations. Do not run tests against systems you are not authorized to test.

10. Account and Access Troubleshooting

1. Verify the user's identity.
2. Confirm the exact resource requested.
3. Check account status using authorized tools.
4. Confirm group membership or effective permission.
5. Verify business need and approval.
6. Apply the minimum required access.
7. Test access.
8. Document the approval and result.

Common Access Symptoms

Symptom	Initial Checks
Password rejected	Account status, reset process, authentication method
Account locked	Approved identity-management status and unlock process
Access denied	Resource, permissions, group membership, approval
Application sign-in failure	Account status, service status, network
Remote access failure	Approved access method, service, network, authorization

11. Printer and Peripheral Troubleshooting

1. Confirm the correct device/printer is selected.
2. Check power and visible status.
3. Check physical connectivity.
4. Check whether other users are affected.
5. Review the local queue where appropriate.
6. Retry with a small test job.
7. Restart approved services/devices only when authorized.
8. Escalate hardware or shared-service failures.

12. Remote Support Troubleshooting

1. Confirm authorization for remote assistance.
2. Confirm the correct target device.
3. Verify the approved remote service is available.
4. Check hostname/address and authorized connectivity.
5. Review relevant service status and logs.
6. Test using an authorized account.
7. Perform only the requested/approved actions.
8. Verify the user's issue is resolved.
9. Disconnect the remote session and document the result.

Linux SSH Example

Purpose	Example
Identify SSH service	<code>systemctl list-units --type=service grep -E 'ssh sshd'</code>
Check service	<code>systemctl status <actual-service-name> --no-pager</code>
Review logs	<code>journalctl -u <actual-service-name> -n 20 --no-pager</code>
Test authorized local session	<code>ssh "\$USER"@localhost</code>

The service name must be discovered first. Do not type a placeholder such as <actual-service-name> literally.

13. Storage and Disk Problems

1. Check available storage.
2. Identify approved temporary/cache data.
3. Check whether a specific application is consuming excessive space.
4. Remove only files permitted by policy.
5. Recheck available capacity.
6. Escalate if critical storage remains low or business data may be affected.

Important

Do not delete unknown business files simply to free space. If data ownership or retention is unclear, escalate.

14. Email and Collaboration Problems

1. Confirm network connectivity.
2. Confirm the correct account is being used.
3. Capture the exact error.
4. Determine whether sending, receiving, synchronization, or calendar functionality is affected.
5. Check approved service status.
6. Test a controlled operation.
7. Escalate suspected service-wide outages or account-security issues.

15. Security-Related Symptoms

Examples

- Unexpected MFA prompts
- Suspicious email or attachment
- Unknown software or pop-ups
- Unexpected account activity
- Suspected malware
- Lost/stolen company device
- Unexpected encryption or data-access behavior

Procedure

1. Treat the issue as potentially security sensitive.
2. Do not open suspicious files or links solely for investigation.
3. Preserve relevant evidence.
4. Follow approved isolation or containment instructions.
5. Escalate to the security/IT administration team.
6. Document timestamps, observed symptoms, and actions.

Do not perform destructive malware-removal actions unless explicitly authorized by the organization's incident-response procedure.

16. Troubleshooting Decision Matrix

Symptom	First-Line Action	Escalate If
No power	Check power/cables/indicators	Hardware failure suspected
No network	Check adapter/configuration/gateway	Multiple users or infrastructure issue
Slow device	Check resources/storage/processes	Persistent or hardware-related
App crash	Capture error/check logs/restart	Recurring or widespread
Access denied	Check resource/permissions/approval	Privileged or authorization issue
Remote access fails	Check approved service/connectivity	Service/firewall/authentication issue
Printer unavailable	Check status/queue/connectivity	Shared service/hardware issue

Suspicious activity	Stop interaction and report	Security incident suspected
Low storage	Check capacity/remove approved temporary data	Data risk or critical capacity

17. Escalation Package

When escalating, provide enough information for the next technician to continue without repeating basic work.

Field	Required Content
Ticket ID	Reference number
User/Department	Affected requester and business unit
Impact	Business effect and scope
Symptom	Exact observed behavior
Timeline	Start time and relevant events
Tests	Checks and results
Changes	Recent changes if known
Actions	Corrective steps already attempted
Evidence	Relevant sanitized logs/screenshots
Escalation Reason	Why first-line support cannot complete the work

18. Resolution Verification

- Repeat the failed action and confirm success.
- Test the relevant user workflow.
- Confirm that the corrective action did not create a new issue.
- Ask the user to validate when appropriate.
- Record the verification result in the ticket.

19. Troubleshooting Documentation Template

Field	Entry
Problem Statement	
Business Impact	
Affected User/Device	
Start Time	
Recent Changes	
Initial Checks	
Evidence	
Hypothesis	
Corrective Action	
Verification	
Root Cause / Likely Cause	
Escalation	
Final Status	

20. Common Troubleshooting Mistakes to Avoid

- Changing multiple variables at once without recording them.
- Restarting or deleting data before understanding the impact.
- Using unapproved diagnostic tools.
- Disabling security controls to bypass an error.
- Making privileged changes without approval.
- Failing to test after a change.
- Closing a ticket without documenting verification.
- Collecting or storing unnecessary sensitive information.

21. Final Troubleshooting Checklist

Control	Complete
Symptom clearly documented	☐
Impact and scope determined	☐
Relevant evidence collected	☐
Recent changes considered	☐
Least-disruptive action used	☐
Security/privacy requirements followed	☐
Corrective action documented	☐
Resolution verified	☐
User validation completed when applicable	☐
Escalation documented when required	☐
Ticket updated with final result	☐

22. Document Maintenance

Review this manual annually and after significant changes to operating systems, applications, network services, support processes, security controls, or escalation responsibilities.

23. Revision History

Version	Date	Description	Owner
1.0	2026-08-10	Initial professional troubleshooting manual for Phase 07.	IT Support Team